

Homework 11

STAT 300 (Fall 2024)

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Clearly label your answers to each question and each sub-question. Your answers MUST be uploaded to Canvas as a <HWx_Yourfirstname.nb.html> file by the deadline.

(1). A researcher is interested in the effect of two different culture media and two different times on the growth of a certain bacteria. Making the runs in a random order, she has collected data on the growth of the bacteria under each factor combination levels. Data can be found in *virus.csv* file.

- (a) Recognize the experimental units, factors of interest and the response variable in this study?
- (b) Use *aggregate()* function to find the means for factor level combinations.
- (c) Visualize the information in an interaction plot. Comment on your observations.
- (d) Fit an *appropriate model* to investigate the relationship between the bacteria growth and factor variable(s) of interest.
- (e) At 5% significance level, test if the effect of culture on the growth depends on the time. Set up appropriate null and alternative hypotheses and complete the test.
- (f) Check the validity of the model assumptions.
- (g) Researcher wants your help with answering below questions. Use R to produce any outputs you need to answer these questions. Using the generated outputs, provide statistical conclusions for the questions.
 - (i) For the bacteria cultivated on medium 1, is there a significant difference in average growth between leaving for 18 hours as oppose to 12 hours?
 - (ii) For the bacteria cultivated on medium 2, is there a significant difference in average growth between leaving for 18 hours as oppose to 12 hours?

(2). Audiologist use standard lists of 50 words for test hearing in which the words were calibrated using subjects with normal hearing to make 50 words in the list equally hard to hear. The goal of the study is to compare four such lists. In the study, 24 subjects has listened to each of the 4 lists in a noisy background. Order in which they listened was randomized. and the percentage of correctly recognized words is provided in the *HearingTest* dataset in *Stat2Data* package.

- (a) Load the data and print the first 3 rows.
- (b) List any factors of interest and any nuisance factors in the study.
- (c) Recognize the study design for the problem.
- (d) Fit a suitable model that would help answer Audiologist's question.
- (e) Set up appropriate hypothesis test that would answer his research question. Based on your fitted model, state the conclusion at 5% significance level.
- (f) Audiologist is further interested in finding answers to below. Use R to produce any outputs you need to answer these questions. Using the generated outputs, provide statistical conclusions for the questions.
 - (i) Lists 1 and 2 were developed recently and 3 and 4 were lists he has been using for a while. Is the average rate of correct recognition significantly different between new and old lists.

- (ii) Is there a significant difference between two lists that were developed recently.
- (3). An experiment was conducted to investigate the effect of the paper color: Blue, Green, Orange on the response rate for questionnaires leave on the windshield of vehicles in supermarket parking lots. A random sample of 15 supermarkets in the area were chosen for the study and each color was assigned at random to five lots. Response rate for the questionnaires were given in the *quest.csv* file.
- Make a side-by-side dotplot of the data and comment on your observations.
 - Fit a one-way anova model to see if there is a significant difference in the average response rate for at least one colour.
 - After the findings in part (b) an executive has made a comment saying that they should have simply printed the questionnaires on white paper. Suppose that you think size of the parking lot might be interfering any relationship there might be between the color of the paper and the average response rate. Use an appropriate graph that display the relationship between response rate and the lot size by the color of the paper. Comment on your observations.
 - Based on your observations, suggest an appropriate model. Write down the model and define all the terms in the context of the problem.
 - Fit the model you suggested in part (d).
 - Based on the fitted model, does it looks like the color of the paper matters? Explain your answer.
- (4). Two types of dementia are dementia with Lewy Bodies and Alzheimer's. To answer this problem you will be using *LewyBody2Groups* data in *Stat2Data* package. The variable Type has two levels: DLB (only Lewy Bodies dementia), DLB/AD (both). Variable MMSE measures the change in functional performance on the Mini Mental State Examination. We are interested in using MMSE to predict whether or not Alzheimers disease is present.
- Load the data and print the first couple of observations.
 - Make an appropriate graph to display the relationship between MMSE and the type of dementia. Comment on your observations.
 - Fit a logistic regression model to this data. For this, you will have to first create a binary variable for the type. Use below code lines for this.

```
#Name too long
Alz<-LewyBody2Groups
Alz$TypeCode<-ifelse(Alz$Type=="DLB",0,1) #You should be using this dataset for remainder of the problem
```

- Based on the fitted model, does MMSE seems to be a significant predictor in determining the probability of Alzheimer's disease (use 5% significance level)? Explain your answer.
- Use the fitted model to estimate the odds of Alzheimer's disease if a patient's MMSE is -4.
- Use the model to estimate the probability of Alzheimer's disease if a patient's MMSE is -4.
- How much does the estimated odds change if the MMSE changes from -4 to -3.